

# Project Roadmap

## OpenAlex AI Research Paper Database Comparison

### Recommended repository name

openalex-ai-db-comparison

### Group members

- Hamza Abdul Kader, matricola 2230150
- Leonardo Marasca, matricola 2217140

### Phase 1: Dataset Collection

- Use the OpenAlex API to collect a small subset of AI, Machine Learning, and RAG-related papers.
- Include papers, authors, topics, abstracts, citations, and publication years; save the raw data as JSON or CSV.

### Phase 2: Data Cleaning

- Remove duplicate and incomplete records, keep only useful fields, and prepare clean files for both databases.

### Phase 3: PostgreSQL Model

- Design relational tables such as papers, authors, topics, citations, paper\_authors, and paper\_topics.
- Load the cleaned data into PostgreSQL and write SQL queries for the selected analyses.

### Phase 4: Neo4j Model

- Design graph nodes such as Paper, Author, and Topic.
- Design relationships such as AUTHORED, CITES, and HAS\_TOPIC; load the same data and write Cypher queries.

### Phase 5: Comparison

- Run equivalent queries in PostgreSQL and Neo4j.
- Compare query complexity, execution time, readability, modeling difficulty, and expressiveness.

### Phase 6: Final Presentation

- Prepare charts and tables showing the comparison results.
- Build slides explaining the dataset, models, queries, results, conclusions, and a short live demo.

### Work division

Hamza will lead OpenAlex data collection, data cleaning, and the PostgreSQL part. Leonardo will lead the Neo4j graph model, Cypher queries, comparison tables, and slides. Both members should understand the full project before the final presentation.